Software Requirements Specification

for

Digital Cake Ordering & Management System.

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Bachelor of Information Technology

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Revision History

Version	Date	Description of Change	Author
1.0	02/12/2025	Initial SRS document created including Introduction, Scope, and Overall Description.	*M D G K Perera*
1.1	03/12/2025	Added Product Features, User Classes, Operating Environment, and Constraints.	*M D G K Perera*
1.2	04/12/2025	Added System Features with detailed requirements, diagrams (DFD, ERD, Class Diagram).	*M D G K Perera*
1.3	05/12/2025	Added all Non-functional Requirements: Safety, Security, Performance, Quality Attributes.	*M D G K Perera*
1.4	06/12/2025	Added Appendices (Glossary, Analysis Models, Issues List). Improved formatting and structure.	*M D G K Perera*
2.0	07/12/2025	Final version reviewed and completed for submission.	*M D G K Perera*

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to clearly define and document all functional and non-functional requirements of the Digital Cake Ordering and Management System. This document establishes a shared understanding between the client, development team, project managers, testers, and all other stakeholders regarding what the system will deliver and how it is expected to operate.

This SRS serves as the foundation for the system's design, implementation, quality assurance, and future maintenance. It specifies the software behaviours, interactions, constraints, assumptions, and performance expectations necessary to ensure that the final system meets the needs of the bakery business and its customers.

The system is intended to support small and medium-sized bakeries by digitizing key business processes, including online cake ordering, cake customization, order tracking, inventory management, customer communication, and administrative reporting. The SRS fully describes the software components of the system, such as:

- Customer-facing online ordering platform
- Order customization and scheduling features
- Administrative dashboard for order and inventory management
- Backend services handling data storage, notifications, payments, and analytics

This document focuses exclusively on the software requirements of the system. It does not include specifications for hardware, network infrastructure, bakery operational procedures, supply chain processes, or any physical activities involved in cake production.

The Purpose section ensures that all stakeholders have a clear and consistent understanding of the system's goals and boundaries before the development process begins.

1.2 Document Conventions

This Software Requirements Specification follows the structure and formatting guidelines of the **IEEE Standard 830** for SRS documentation. The following conventions are used throughout this document to ensure clarity, uniformity, and ease of understanding:

1.2.1 Text Formatting Conventions

- **Bold text** is used for section headings, key terms, and important concepts.
- *Italic text* is used to emphasize specific words, definitions, or notes.

- **Monospaced text** is used for identifiers, system messages, requirement codes, and data fields.

1.2.2 Requirement Identification Conventions

To maintain consistency and traceability, all requirements are uniquely labelled using the following prefixes:

- **FR-x** → Functional Requirements
- **NFR-x** → Non-Functional Requirements
- **BR-x** → Business Rules
- **UI-x** → User Interface Requirements
- **DB-x** → Database-Related Requirements
- **UC-x** → Use Cases

Each requirement is assigned:

- A unique identifier
- A priority: **High (H)**, **Medium (M)**, or **Low (L)**
- A clear and testable description

1.2.3 Diagram Conventions

All diagrams included in this document follow standard notations

- **Use Case Diagrams** – UML Notation
- **ER Diagrams** – Crow's Foot Notation
- **DFD Level 0 & Level 1** – Gane & Sarson symbols
- **Sequence & Activity Diagrams** – UML Notation
- **Flowcharts** – Standard ANSI symbols

Diagrams included in the SRS may be represented using **Mermaid notation** for digital readability.

1.2.4 Terminology Conventions

- The term **"system"** refers to the Digital Cake Ordering and Management System unless otherwise stated.
- The term **"user"** refers to any individual interacting with the system, including customers and administrators.
- The term **"admin"** refers specifically to bakery staff or system managers.

1.2.5 Priority Conventions

Priority of requirements is categorized as:

- **High (H)** – Critical for system operation; must be included in Release 1.0
- **Medium (M)** – Important but may be implemented in later phases
- **Low (L)** – Optional enhancements or future upgrades

1.3 Intended Audience and Reading Suggestions

The intended audience for this document includes the **module lecturer, examiners, and any academic reviewers** who will evaluate the design and completeness of the Digital Cake Ordering and Management System. Although this project is completed individually, this SRS is written following industry standards to demonstrate proper documentation practices.

This document will serve as a structured reference for understanding the system's purpose, requirements, and expected behaviour. It outlines all functional and non-functional requirements in a clear and organized manner, which will assist reviewers in assessing the correctness, clarity, and completeness of the proposed system.

To ensure that this SRS is easily readable and understandable, it has been organized into sections and subsections. Each section includes a concise heading and description summarizing its contents. The requirements of the Digital Cake Ordering and Management System have been presented in a consistent format, including purpose, description, and criteria where applicable. Diagrams such as context diagrams, DFDs, use case models, and ER diagrams have been included to enhance clarity and provide visual representations of system processes.

Readers are advised to begin with the introduction section, which provides an overview of the project and defines the purpose of the SRS. After understanding the background, readers may proceed to the overall description section to gain insight into the system's scope and general operation. From there, the system features section provides detailed descriptions of all functional requirements relevant to the system's operation. Additional sections, such as external interfaces and non-functional requirements, should be reviewed to understand the system's technical expectations and constraints.

Since this is an academic project completed independently, the document focuses primarily on the system requirements rather than stakeholder collaboration. Nevertheless, the acceptance criteria and defined requirements should be reviewed carefully to ensure they align with the intended outcomes of the project.

1.4 Project Scope

The Digital Cake Ordering and Management System provides an online platform that allows users to browse cakes, customize their selections, and place orders efficiently. The system also includes administrative functionalities for managing orders, monitoring inventory, and updating product details. Although developed as an individual academic project, the system models real-world operations commonly found in small to medium-sized bakeries. The primary aim of this system is to streamline the cake-ordering workflow, improve accessibility, and provide a smoother user experience through automation.

This document outlines the requirements, constraints, and expectations necessary to define and understand the intended system. The scope is limited to the software components implemented for academic demonstration and does not include real commercial integrations.

1.4.1 System Goals

The main goals of the Digital Cake Ordering and Management System are as follows:

- To provide a user-friendly online interface for customers to view and order cakes.
- To support customization of cakes based on size, flavour, decorations, and special instructions.
- To offer administrators tools to efficiently manage incoming orders and inventory levels.
  To build a functional prototype that demonstrates key concepts of software engineering, user interaction design, and requirement documentation.

1.4.2 System Objectives

The specific objectives of the system include:

1. **Enable Online Ordering:**
   Allow users to browse the available cake menu and submit orders directly through the web interface.
2. **Support Cake Customization:**
   Provide customization options (size, type, flavour, toppings, icing, messages, etc.) before an order is placed.
3. **Manage Inventory:**
   Maintain ingredient stock levels and update them automatically based on order quantities.
4. **Administrator Dashboard:**
   Provide an admin interface to update cake details, manage inventory, and process customer orders.

1.5 References

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2. Overall Description

This section provides a comprehensive overview of the Digital Cake Ordering and Management System. It explains the system's environment, its major functional components, the types of users involved, the operating constraints, and the assumptions upon which the system is based. This high-level description helps readers understand the context of the system before examining the specific requirements in later sections.

2.1 Product Perspective

The Digital Cake Ordering and Management System is a self-contained, web-based application developed as a standalone academic project. It is not part of a larger system, nor does it rely on external bakery management software. Instead, it simulates real-world functionality commonly found in online bakery ordering platforms.

The system is designed using a **three-tier architecture**, consisting of:

2.1.1 Presentation Layer (Frontend)

- Provides the user interface for customers and admin users.
- Handles cake browsing, customization forms, order submission, and dashboard interactions.
- Designed to be accessible through a standard web browser.

2.1.2 Application Layer (Backend Server)

- Handles logic for processing orders, validating inputs, saving data and managing product listings.
- Facilitates communication between the frontend and the database.
- Implements business rules such as order status updates and ingredient usage calculations.

2.1.3 Data Layer (Database)

- Stores persistent data including products, orders, user information (if login is included), and inventory levels.
- Ensures secure and consistent access to stored information.

2.1.4 External Interfaces

Although the project is implemented individually, the system design includes optional placeholders for:

- Email/SMS notification services
- API connections for future payment integration

- Reporting modules

System Context Diagram

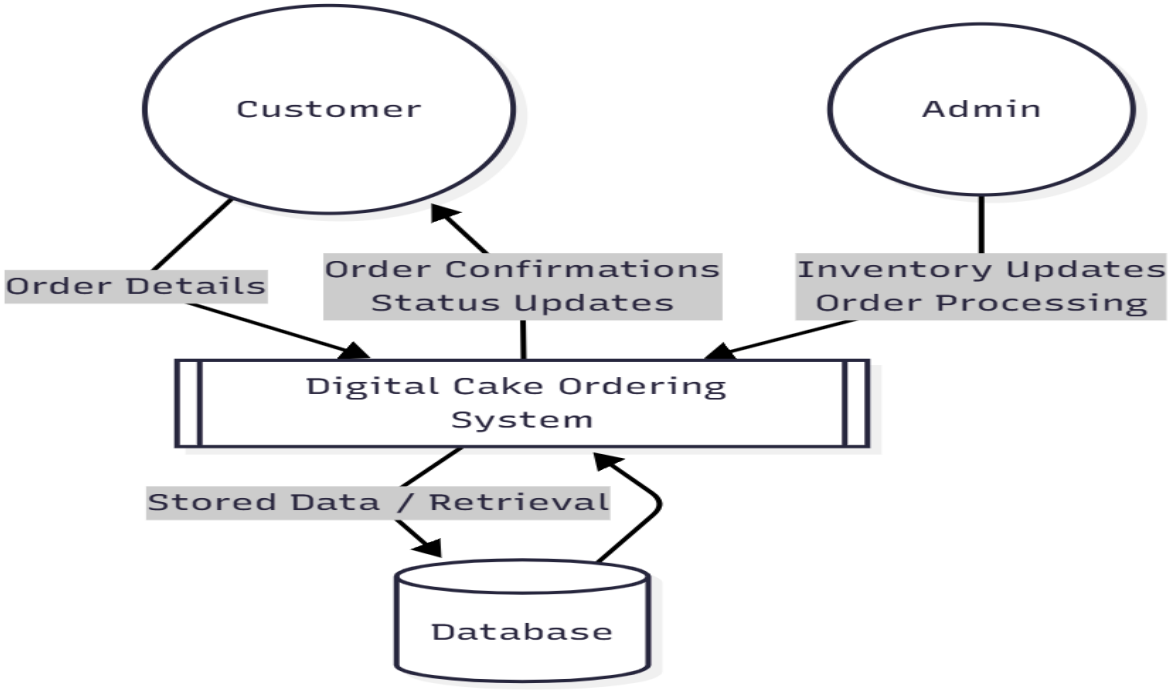

Figure 1 System Context Diagram

2.2 Product Features

The Digital Cake Ordering and Management System provides a set of interconnected features that allow users to browse cakes, customize their selections, place orders, and enable the administrator to manage inventory and order workflows. This section summarizes the major functional capabilities of the system. Detailed descriptions of each feature will be provided in Section 3 – System Features.

1. Cake Browsing

This feature enables customers to explore all cakes available in the system. The cake catalogue presents each item with essential product information to help users make informed ordering decisions.

Details include:

- Display of cake name, image, description, and price
- Optional categorization (e.g., chocolate cakes, birthday cakes, wedding cakes)
- Filtering options such as flavour, size, or price range
- Search function to quickly locate specific cakes

- Ability to view detailed cake information including available customization options

This feature ensures that customers have a clear and engaging browsing experience.

2. Cake Customization

The system provides a customization interface that allows customers to personalize their selected cake according to their preferences.

Customization options may include:

- Cake size (small, medium, large, etc.)
- Flavour selection (vanilla, chocolate, red velvet, etc.)
- Icing/frosting type
- Decorations (sprinkles, fondant, fruit toppings, etc.)
- Custom message or greeting text on the cake
- Additional special instructions (e.g., reduce sugar, add candles)

The feature ensures flexibility and helps customers design cakes suitable for their events or preferences.

3. Online Order Placement

This feature allows customers to place their cake orders through a structured online process.

Order placement includes:

- Capturing customer information (name, phone number, address if delivery is used)
- Selecting pickup or delivery date and time
- Review of selected cake items and quantities
- Display of total price and breakdown of costs
- Submitting the order for processing by the administrator

This feature eliminates the need for phone or in-person ordering, improving convenience and accuracy.

4. Order Tracking

After placing an order, customers can monitor the progress of their order through a clear status update system.

Order statuses may include:

- **Pending:** Order received but not processed
- **Accepted:** Admin has acknowledged the order
- **In Progress:** Cake preparation has started
- **Ready:** Order is ready for pickup/delivery
- **Completed:** Customer has received the order
- **Cancelled (if applicable):** Order cancelled by admin

This feature enhances transparency and reduces the need for customers to call the bakery for updates.

5. Administrator Order Management

This feature provides the administrator with tools to manage all aspects of customer orders.

Admin capabilities include:

- Viewing all incoming orders with detailed customer and product information
- Updating order status (Pending → In Progress → Ready → Completed)
- Filtering and sorting orders by date, status, or customer
- Viewing special instructions and customization requests
- Managing urgent orders or cancellations
- Accessing an order history database

This feature ensures efficient workflow management for bakeries or system owners.

6. Inventory Management

The system allows the administrator to maintain and update ingredient stock levels to ensure that the bakery can fulfil customer orders.

Inventory features include:

- Manual update of ingredient quantities
- Viewing current stock levels of cake components (flour, sugar, icing, flavours, decorations, etc.)
- Ingredient usage updates based on cake orders (optional, depending on implementation)
- Low-stock alerts (if included in your project scope)

This feature helps avoid ingredient shortages and supports planning for restocking.

7. Product Management

Administrators have full control over the cake catalogue displayed to customers.

Product management includes:

- Adding new cakes to the catalogue
- Editing existing cake details (name, price, description, image, availability, etc.)
- Removing cakes that are no longer offered
- Updating customization options available for each cake

This ensures that the system remains up to date with the bakery's actual offerings.

8. Reporting (Optional)

Depending on the academic scope of the project, the system may include basic reporting tools.

Possible reports include:

- Number of orders received daily/weekly/monthly
- List of most popular cakes
- Total revenue (if implemented)
- Inventory usage summaries
- Pending vs. completed orders statistics

Reports help the administrator review business performance and make informed decisions.

Class Diagram

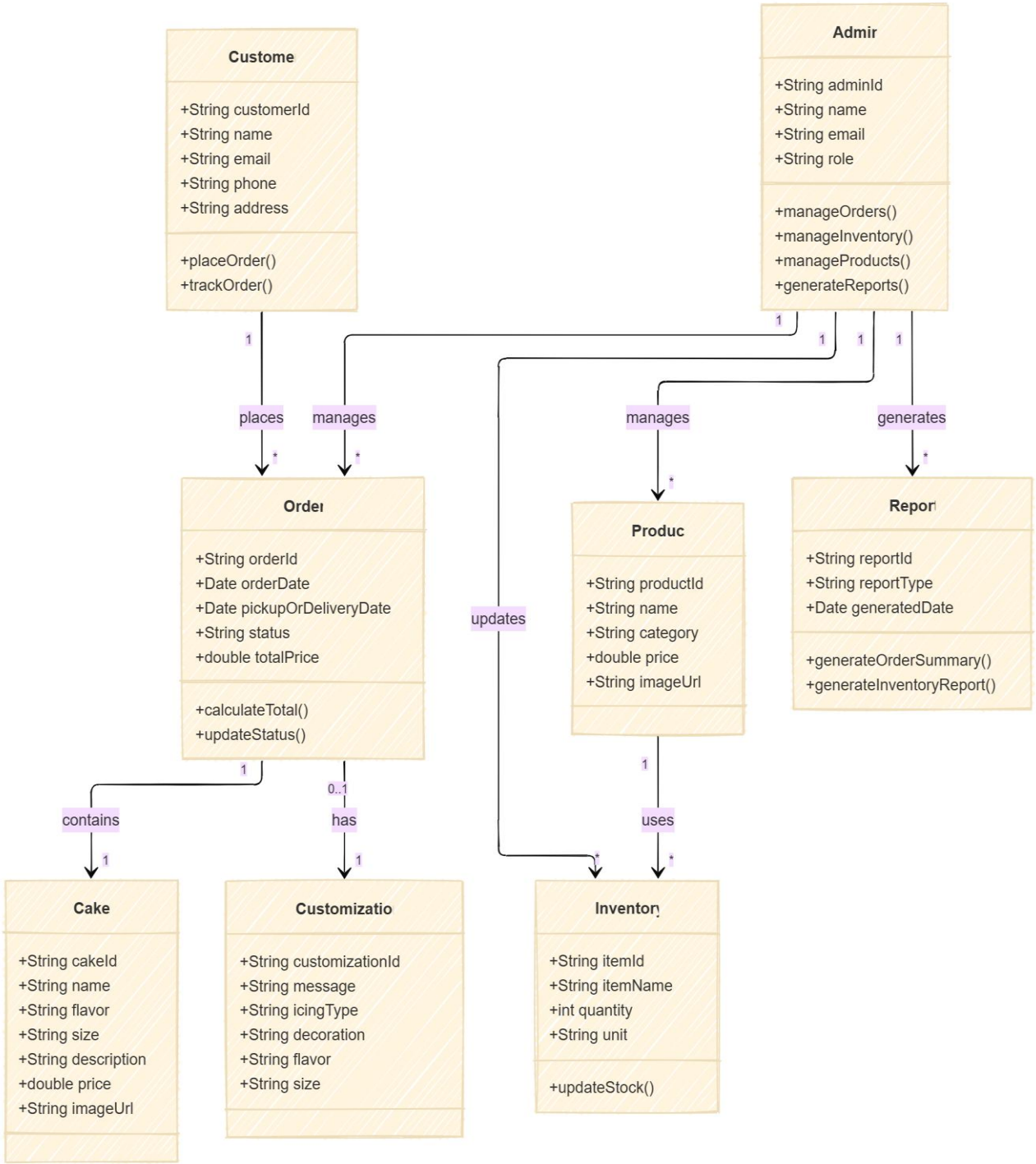

Figure 2 Class Diagram

2.3 User Classes and Characteristics

The Digital Cake Ordering and Management System is designed to support two primary user classes. These classes differ in their system access levels, frequency of system usage, technical expertise, and functional requirements. A summary of each class is provided below.

1. Customer (Primary User – External)

Description:

- Individuals who wish to browse cakes, customize orders, and place online cake orders. Customers interact only with the front-facing web interface.

Characteristics:

- **Technical Expertise:** Basic internet and mobile device usage skills.
- **Frequency of Use:** Occasional; may use the system during special events or order intervals.
- **Access Level:** Lowest privilege; restricted to browsing, customizing, placing orders, and tracking order status.
- **Usage Motivation:** Convenience, online ordering, customization features.

Functions Used:

- Browse cake catalogue
- Customize cake attributes
- Place online orders
- Track real-time order status

Special Considerations:

- Interface must be user-friendly, visually appealing, and easy to navigate.
- Error messages must be clear and non-technical.
- No training required.

2. Administrator (Internal User – Bakery Owner / Staff)

Description:

- Administrative personnel responsible for managing backend operations such as orders, inventory, and products.

Characteristics:

- **Technical Expertise:** Moderate familiarity with web-based management panels.
- **Frequency of Use:** High; used daily for order processing and inventory adjustments.
- **Access Level:** Highest privilege; full access to administrative modules.
- **Usage Motivation:** Workflow management, business tracking, inventory monitoring.

Functions Used:

- View and manage orders
- Update order statuses
- Manage inventory items and stock levels
- Add, edit, or remove cake products
- View basic reports (optional)

Special Considerations:

- Requires secure login authentication.
- System should support efficient workflows to reduce processing time.
- Training may be minimal, but system must remain intuitive for frequent use.

User Importance Ranking

User Class	Priority	Reason
Customer	High	Generates revenue; essential for system purpose
Administrator	High	Ensures business operations run smoothly; supports fulfilment

Both user classes are considered **equally critical** because the system's success depends on smooth interactions between customer ordering and administrative processing.

2.4 Operating Environment

The Digital Cake Ordering and Management System will operate in a modern web-based environment, ensuring compatibility, accessibility, and ease of deployment for both customers and administrators. The environment includes client-side requirements, server-side infrastructure, supported technologies, and third-party components required for smooth system operation.

1. Client-Side Environment (Customers & Administrators)

Hardware Requirements

- Any modern computing device such as:
  - Desktop or laptop computer
  - Smartphone (Android/iOS)
  - Tablet device
- Minimum resolution: **1280 × 720** (desktop), **360 × 640** (mobile)

Operating Systems

- Windows 10 or later
- macOS 11 or later
- Android 9 or newer
- iOS 13 or newer

Supported Web Browsers

The system is fully browser-based and supports:

- Google Chrome (latest two versions)
- Mozilla Firefox (latest two versions)
- Microsoft Edge (latest version)
- Safari (for iOS/macOS, latest two versions)

Other Client Requirements

- Stable internet connection
- JavaScript enabled
- Cookies enabled for session handling

2. Server-Side Environment

Operating System

The system backend may be hosted on any of the following server environments:

- Linux (Ubuntu 20.04+ recommended)
- Windows Server 2019 or later

Backend Technologies

- **Node.js (v16+)** – for server-side logic and API processing

- **Express.js** – for handling RESTful routes
- **MongoDB (v5+)** – for database storage

Web Server

- NGINX or Apache (optional reverse proxy)

Runtime Environment

ECMAScript 6+ compatible environment

3. Software Dependencies and Frameworks

Frontend

- HTML5, CSS3
- JavaScript
- React.js (v18+)
- Axios or Fetch API (for API communication)

Backend / Database

- MongoDB Driver / Mongoose ORM
- JSON Web Token (JWT) for authentication
- Node mailer (optional email notifications)

Optional Third-Party Integrations

- Payment gateway APIs (if implemented)
- Email/notification services
- Cloud storage (AWS S3, Firebase Storage)

4. Interoperability Requirements

The system must coexist peacefully with:

- Existing web hosting environments
- SSL certificates for secure HTTPS access
- Standard HTTP/HTTPS communication protocols
- JSON-based RESTful APIs
- Backup services or database replication tools

5. Deployment Environment

The system supports the following deployment models:

- **Local deployment** for academic use
- **Cloud deployment** (AWS EC2, Heroku, Vercel, Render)
- **On-premises hosting** at the bakery (if chosen)

6. Constraints

- Internet connectivity is required for both customer and admin operations
- Device must support modern browser features (ES6, DOM API, etc.)
- Backend server must remain available for real-time order status updates

2.5 Design and Implementation Constraints

The design and implementation of the Digital Cake Ordering and Management System must comply with specific technical, security, and operational constraints. These limitations influence technology selection, system architecture, integration capabilities, and development scope.

2.5.1 Technology and Platform Constraints

1. Mandatory Technology Stack

The system must use the following technologies as part of the project requirements:

- **Backend Framework:** Spring Boot (Java)
- **Frontend Framework:** React.js (JavaScript)
- **Build/Utility Platform:** Node.js (used for React build, package management, and UI dependencies)
- **Database:** PostgreSQL (Relational DBMS)
- **API Type:** RESTful JSON APIs

These constraints restrict developers from using:

- Alternative backend languages (PHP, Python, JavaScript-only, etc.)
- NoSQL databases like MongoDB
- Server-side rendering frameworks

2.5.2 Hardware and Performance Constraints

- Server must support Java 17+ for Spring Boot runtime.
- PostgreSQL requires sufficient memory allocation for relational queries.

- The system must operate efficiently on:
  - Windows/macOS/Linux machines for development
  - Standard cloud hosting or on-premises servers for deployment
- Mobile devices with limited RAM must still load client UI efficiently.

2.5.3 Deployment Constraints

- Backend must be deployable on **Tomcat embedded server** through Spring Boot.
- PostgreSQL must be installed and configured on the hosting environment.
- System must run in a **multi-tier architecture** (frontend–backend–database).
- HTTPS must be used for all external communication.

2.5.4 Security and Privacy Constraints

- Data must be encrypted in transit using **TLS/SSL**.
- Passwords must be hashed using secure algorithms (e.g., BCrypt).
- Spring Security (or equivalent) must enforce:
  - Authentication (Admin access)
  - Role-based authorization
- PostgreSQL must follow proper privacy practices:
  - No storage of raw payment data
  - No storing sensitive data beyond project needs

2.5.5 UI/UX Design Constraints

- React must follow component-based structure with Hooks (no class components unless required).
- UI must be responsive and mobile-friendly.
- Node.js and NPM packages must adhere to compatibility with React 18+.
- Limited use of heavy UI frameworks due to performance constraints.

2.5.6 Integration Constraints

- Backend and frontend must communicate via REST (JSON format).
- PostgreSQL schema must be relational and normalized.
- System must support integration only through REST APIs; SOAP or other protocols are not allowed.
- Optional integrations (e.g., email service, payment gateway) must support HTTP-based APIs.

2.5.7 Coding and Architectural Standards

- Backend must follow Spring Boot conventions:

  - Controller → Service → Repository layered architecture
  - JPA/Hibernate for ORM with PostgreSQL
  - Standard DTOs and Entities
- React must follow:
  - Functional components
  - React Router for navigation
  - State handled using React hooks or Context API
- Node.js used strictly for UI dependencies (npm, webpack, Vite, etc.)

2.5.8 Time and Resource Constraints

- Academic project scope limits:
  - Large-scale testing
  - High-level scalability features
  - Full enterprise-level security mechanisms
- Only essential features will be implemented due to time limitations.

2.5.9 Maintenance Constraints

- The system must remain maintainable by a single developer.
- Database schema changes must be backward-compatible.
- Code must be well-documented for future enhancements.

2.6 User Documentation

The Digital Cake Ordering and Management System will include a set of user documentation materials designed to assist both customers and administrators in understanding and effectively using the system. The documentation ensures usability, reduces training time, and supports the maintenance of the system.

2.6.1. Documentation for Customers

These materials support end-users interacting with the cake ordering interface.

1.1 Online User Guide

- Step-by-step instructions for browsing cakes, customizing orders, and completing checkout.
- Accessible directly through the system's "Help" or "Support" section.
- Delivered as **HTML pages** for mobile and desktop compatibility.

1.2 Quick Start Guide

- A simplified one-page guide showing how to place an order in 3–5 steps.

- Included as:
  - Downloadable **PDF**
  - Displayed as an introductory popup for first-time users

1.3 FAQ Section

- Covers common customer questions such as:
  - How to customize cakes
  - Delivery vs. pickup
  - Payment-related instructions (if implemented)
- Delivered in interactive **web format**.

2.6.2. Documentation for Administrators

Admin documentation supports bakery staff in handling the backend tasks.

1. **Administrator Manual**

- Detailed instructions to manage:
  - Orders
  - Inventory
  - Products
  - Reporting features
- Delivered as a downloadable PDF and a web-based manual.

2. **Training Videos (Optional)**

- Short screen-recorded tutorials demonstrating:
  - Order management workflow
  - Inventory update process
  - Product addition/editing
- Delivered in MP4 format or embedded in the Admin Help section.

3. **Tooltips and Inline Help**

- Context-sensitive assistance built into the UI.
- Hover-over tips and mini popups explaining form fields, buttons, and dashboard features.

2.6.3. Developer / Technical Documentation (Optional for Academic Project)

If included, the documentation will help future developers maintain or enhance the system.

1. **API Documentation**

- REST API reference for backend endpoints.

- Delivered via:
    - **Swagger/OpenAPI** (if implemented)
    - PDF or Markdown files

2. **System Architecture Guide**

- Explains:
    - Spring Boot project structure
    - PostgreSQL schema
    - React frontend structure
- Delivered as PDF/Markdown.

2.6.4. Documentation Delivery Format Standards

- All downloadable documents will be provided in **PDF format**, following standard A4 layout.
- In-app help sections will be delivered as:
    - HTML5 pages
    - Responsive and mobile-friendly layouts
- Video tutorials (if included) will follow **MP4** format with standard resolution (720p or higher).
- All text documentation will follow:
    - Clear headings and subheadings
    - Stepwise instructions
    - Screenshots (if available)
    - Consistent terminology across all materials

2.7 Assumptions and Dependencies

This section identifies the assumptions made during the requirements definition process and the dependencies that may influence the development, deployment, and operation of the Digital Cake Ordering and Management System. Any change in these assumptions or dependencies may impact system functionality, performance, or feasibility.

2.7.1 Assumptions

1. User and Operational Assumptions

- Users (customers and administrators) will have stable internet access during system usage.
- Customers possess basic computer or smartphone literacy required to navigate a web interface.

- Administrators will have sufficient knowledge to use the system after minimal training.
- The bakery will manually update inventory data, and the system will not auto-detect ingredient usage.

2. Technical Assumptions

- The backend server environment will support:
  - Java 17+
  - Spring Boot 3+
- The frontend environment will support:
  - Node.js 16+
  - React 18+
- PostgreSQL will be properly installed, configured, and accessible from the backend.
- HTTPS will be available for secure communication.
- All devices accessing the system will support modern browsers with JavaScript enabled.

3. Development Assumptions

- The project will be developed and maintained by a single developer or a small development team.
- Required development tools (IDE, database client, NPM, build tools) will remain available throughout the project lifecycle.
- Any optional integration (email services, cloud storage, payment gateway) will have stable APIs and documentation.
- Features are implemented within the scope and time limits of an academic project.

2.7.2 Dependencies

1. Software Dependencies

The system depends on the proper functioning of the following technologies:

- **Spring Boot (Backend):**
  Required for handling business logic, security, and REST APIs.
- **React.js (Frontend):**
  Required for rendering user interface components.
- **Node.js & NPM:**
  Required for building the React application and bundling frontend dependencies.
- **PostgreSQL Database:**
  Required for storing all system data such as orders, users, products, and inventory.
- **Hibernate / JPA:**
  Required for ORM and data persistence in Spring Boot.

2. Third-Party Services (Optional)

The system may optionally depend on:

- Email notification service (e.g., Gmail SMTP, SendGrid, Mailgun)
- Cloud hosting providers (AWS, Azure, DigitalOcean)
- File storage or image hosting (optional)

If these services change APIs or pricing, system behavior may be affected.

3. Browser and Device Dependencies

- System relies on user browsers supporting HTML5, CSS3, ES6 JavaScript, and Fetch/Axios API.
- Mobile responsiveness depends on device screen size and browser version.

4. Security Dependencies

- System security depends on:
  - Correct configuration of HTTPS certificates
  - Proper password hashing (BCrypt)
  - Valid JWT authentication tokens
- Any misconfiguration may lead to security vulnerabilities.

5. Hosting and Deployment Dependencies

- Backend and database hosting must provide sufficient performance for handling user traffic.
- Deployment environment must support:
  - Java runtime
  - PostgreSQL access
  - Node.js build steps

Availability of these services directly affects system uptime.

2.7.3 Risk of Changing Assumptions

If any assumption fails or dependencies change, risks include:

- Reduced system performance
- Communication failures between frontend and backend
- Data loss or corruption
- Inability to authenticate users securely
- Reduced compatibility with user devices
- Delays in project delivery

3. System Features

This section describes the major functional features of the Digital Cake Ordering and Management System. Each feature is organized into descriptive components, stimulus/response behaviours, and detailed functional requirements.

3.1 Cake Browsing

3.1.1 Description and Priority

This feature allows customers to browse a catalogue of available cakes with details such as name, price, flavour, size, and an image.
Priority: High (core functionality for customer ordering)

3.1.2 Stimulus/Response Sequences

- **Stimulus:** Customer visits the website or product catalogue page.
- **Response:** System displays a list of cakes with images and basic details.
- **Stimulus:** Customer clicks a specific cake.
- **Response:** System displays detailed cake information.

3.1.3 Functional Requirements

- **REQ-BR-01:** The system shall allow users to view a list of all available cakes.
- **REQ-BR-02:** The system shall display cake details including name, flavour, size options, price, and image.
- **REQ-BR-03:** The system shall allow users to filter cakes by category, flavour, or price (if implemented).
- **REQ-BR-04:** The system shall update the catalogue automatically when products are added/edited by the admin.

3.2 Cake Customization

3.2.1 Description and Priority

Customers can personalize cakes by selecting options such as size, flavour, icing, decorations, and custom messages.
Priority: High

3.2.2 Stimulus/Response Sequences

- **Stimulus:** Customer selects the “Customize” option on a cake.

- **Response:** System displays customization fields and selectable attributes.
- **Stimulus:** Customer saves customization.
- **Response:** System applies customization and updates order summary.

3.2.3 Functional Requirements

- **REQ-CZ-01:** The system shall allow users to select cake size, flavour, and icing options.
- **REQ-CZ-02:** The system shall allow users to enter a custom message for the cake.
- **REQ-CZ-03:** The system shall update the price based on selected customization options.
- **REQ-CZ-04:** The system shall validate customization fields (e.g., message character limit).

3.3 Online Order Placement

3.3.1 Description and Priority

Users can place cake orders by submitting customer details and customization preferences.
Priority: High

3.3.2 Stimulus/Response Sequences

- **Stimulus:** Customer clicks “Place Order”.
- **Response:** System displays an order form.
- **Stimulus:** Customer submits form.
- **Response:** System saves order and generates confirmation.

3.3.3 Functional Requirements

- **REQ-OP-01:** The system shall allow users to submit an online order.
- **REQ-OP-02:** The system shall require customer details including name, phone, email, and address (if delivery).
- **REQ-OP-03:** The system shall allow users to choose a pickup or delivery date.
- **REQ-OP-04:** The system shall generate a unique order ID for each order.
- **REQ-OP-05:** The system shall send an order confirmation message (email optional).

3.4 Order Tracking

3.4.1 Description and Priority

Customers can track real-time status updates of their orders.
Priority: Medium–High

3.4.2 Stimulus/Response Sequences

- **Stimulus:** Customer selects “Track Order”.
- **Response:** System displays current status such as Pending, In Progress, Ready, Completed.

3.4.3 Functional Requirements

- **REQ-TR-01:** The system shall display the status of customer orders.
- **REQ-TR-02:** The system shall update order status based on admin actions.
- **REQ-TR-03:** The system shall show estimated completion time (optional).

3.5 Administrator Order Management

3.5.1 Description and Priority

Admins can review, update, process, and manage customer orders.
Priority: High

3.5.2 Stimulus/Response Sequences

- **Stimulus:** Admin logs into the dashboard.
- **Response:** System displays all incoming and active orders.
- **Stimulus:** Admin updates status.
- **Response:** System saves update and notifies customer (optional).

3.5.3 Functional Requirements

- **REQ-OM-01:** The system shall allow admins to view all orders.
- **REQ-OM-02:** The system shall allow admins to update order status.
- **REQ-OM-03:** The system shall allow admins to view customer details for each order.
- **REQ-OM-04:** The system shall allow admins to filter orders by status.

3.6 Inventory Management

3.6.1 Description and Priority

Admins can update ingredient stock levels and view inventory usage.
Priority: Medium

3.6.2 Stimulus/Response Sequences

- **Stimulus:** Admin selects “Inventory Management”.
- **Response:** System displays ingredient list with quantities.
- **Stimulus:** Admin edits stock value.

- **Response:** System updates inventory records.

3.6.3 Functional Requirements

- **REQ-IM-01:** The system shall allow admins to view all inventory items.
- **REQ-IM-02:** The system shall allow admins to update ingredient quantities.
- **REQ-IM-03:** The system shall restrict inventory access to admin users only.

3.7 Product Management

3.7.1 Description and Priority

Admins can add, edit, or remove cakes displayed in the catalogue.
Priority: High

3.7.2 Stimulus/Response Sequences

- **Stimulus:** Admin selects “Add Cake”.
- **Response:** System displays product creation form.
- **Stimulus:** Admin edits product.
- **Response:** System updates catalogue.

3.7.3 Functional Requirements

- **REQ-PM-01:** The system shall allow admins to add new cake products.
- **REQ-PM-02:** The system shall allow admins to edit existing products.
- **REQ-PM-03:** The system shall allow admins to delete products from the catalogue.
- **REQ-PM-04:** The system shall validate product details before saving.

3.8 Reporting (Optional)

3.8.1 Description and Priority

The system may generate reports for orders and inventory.
Priority: Low to Medium (optional academic feature)

3.8.2 Stimulus/Response Sequences

- **Stimulus:** Admin selects “Reports”.
- **Response:** System displays available summary reports.

3.8.3 Functional Requirements

- **REQ-RP-01:** The system should generate daily or weekly order summaries.
- **REQ-RP-02:** The system should provide a list of best-selling cakes.
- **REQ-RP-03:** The system should generate inventory usage statistics.

4. External Interface Requirements

4.1 User Interfaces

This section describes the logical characteristics, layout rules, standards, and interaction expectations of the user interfaces (UI) used by customers and administrators. Final UI layouts, colours, themes, and screen wireframes will be documented separately in a dedicated User Interface Specification.

4.1.1 General User Interface Requirements

- All interfaces shall follow responsive web design, ensuring compatibility with mobile, tablet, and desktop devices.
- The system shall maintain a consistent visual theme across all pages, including:
    - Standard header with logo and navigation
    - Footer with contact information and links
    - Consistent colour palette and typography
- All screens shall use clear labels, readable fonts, and intuitive icons.
- All error messages shall be displayed in red text and positioned near the field causing the error.
- Required fields shall be marked with an asterisk (*).
- All buttons shall follow a standard style (e.g., rounded corners, hover effect).
- The system shall support simple, linear workflows to minimize user confusion.

4.1.2 Customer User Interface

4.1.2.1 Home Page

- Displays featured cakes and promotional items.
- Provides navigation links to:
    - Cake catalogue
    - Track Order
    - Contact/Help

4.1.2.2 Cake Catalogue Interface

- Displays cake list in grid format.
- Each card shows image, name, price.
- Clicking a cake opens the Cake Detail page.
- Optional filters (Category, Price, Flavour).

4.1.2.3 Cake Detail & Customization Page

- Shows:
    - Large cake image
    - Description
    - Customization fields (size, flavour, icing, custom message)
- “Add to Order” button visible and accessible.
- Clear price updates based on customization.

4.1.2.4 Order Placement Form

- Fields:
    - Name
    - Phone
    - Email
    - Address (if delivery)
    - Pickup/delivery date
- Includes order summary panel.
- Validates required fields and displays errors inline.

4.1.2.5 Order Tracking Interface

- Input: Order ID
- Displays status:
    - Pending
    - In Progress
    - Ready
    - Completed
- Shows optional estimated time.

4.1.3 Administrator User Interface

4.1.3.1 Login Page"

- Username and password fields
- Invalid credentials display a clear error message
- “Forgot Password” link (optional)

4.1.3.2 Admin Dashboard

- Overview cards for:
  - Total Orders Today
  - Pending Orders
  - Low Inventory Items
- Left-side navigation menu for all admin modules.

4.1.3.3 Order Management Screen

- Table showing:
  - Order ID
  - Customer Name
  - Date
  - Status
- Ability to update order status using a dropdown.
- "View Details" option to open full order record.

4.1.3.4 Inventory Management Screen

- Editable table of ingredients and quantities.
- Buttons:
  - Add Item
  - Update Quantity
  - Remove Item
- Warns when item quantity is low.

4.1.3.5 Product Management Screen

- Allows:
  - Add new cake product
  - Edit existing cake
  - Delete product
- Form fields validated before saving.

4.1.3.6 Reports Screen (Optional)

- Dropdown to choose report type.
- Table or graph-based report preview.
- "Download PDF" option (optional).

4.1.4 Error Message and Notification Standards

- All error notifications must be displayed:

  - Near the affected form field
  - In red text
  - With clear explanation (e.g., “Invalid email”, “Stock cannot be negative”)
- Success messages must appear in green.
- System warnings (e.g., low inventory) must appear in amber/yellow.

4.1.5 Accessibility Requirements

- Minimum font size: 14px
- Buttons must be large enough for mobile tap targets
- Colour contrast must meet WCAG accessibility guidelines
- Images must have alt-text attributes

4.2 Hardware Interfaces

This section describes the logical and physical characteristics of the hardware components that interact with the Digital Cake Ordering and Management System. Since the system is primarily web-based, hardware interfaces are minimal and mainly related to user devices, servers, and hosting infrastructure.

4.2.1 Client-Side Hardware Interfaces

Supported Device Types

The system must operate correctly on the following hardware devices:

- **Desktop computers**
  - Intel/AMD-based PCs
  - Minimum 4GB RAM
  - Screen resolution 1280×720 or higher
- **Laptops**
  - Windows, macOS, or Linux
  - Minimum 4GB RAM
- **Mobile devices (Smartphones)**
  - Android (9.0+)
  - iPhone (iOS 13+)
  - Minimum resolution 360×640
  - Touchscreen support required
- **Tablets**
  - Android tablets
  - iPads

- Responsive UI support required

Interaction Characteristics

- Users interact with the system purely through web browsers, requiring no specialized hardware.
- Touch input must be supported for mobile and tablet devices.
- No installation or client-side hardware drivers are necessary.

4.2.2 Server-Side Hardware Interfaces

Backend Server Requirements

The backend built using **Spring Boot** will run on a server with the following minimum specifications:

- **Processor:** Dual-core CPU (Intel/AMD equivalent)
- **Memory:** Minimum 4GB RAM (8GB recommended for production)
- **Storage:** 20GB minimum (SSD recommended)
- **Network:** Stable internet connection with public IP

Interaction Characteristics

- The server communicates with the database server using TCP/IP.
- REST API endpoints handle HTTP(S) requests from client devices.
  No direct hardware-to-software control interfaces exist.

4.2.3 Database Server Hardware Interface

PostgreSQL Database Requirements

- **Processor:** Dual-core CPU
- **Memory:** Minimum 2GB RAM
- **Storage:** Sufficient disk space for database tables and logs
- **I/O:** High read/write performance recommended (SSD)

Interface Characteristics

- Communication between Spring Boot backend and PostgreSQL occurs over:
  - **Port 5432 (default)**
  - **TCP/IP protocol**
- The database server must support:
  - JDBC driver for PostgreSQL

- Network authentication and credential validation

4.2.4 Hosting and Deployment Hardware Environment

The system may be hosted in one of the following environments:

Cloud Hosting (Preferred)

- AWS EC2, Azure VM, DigitalOcean Droplets, Render, etc.
- Virtual hardware specifications dependent on hosting plan.
- Supports horizontal or vertical scaling if traffic increases.

Local On-Premises Server (Optional for small bakeries)

- Requires a local machine running the backend, frontend build, and PostgreSQL database.
- May require UPS backup to avoid downtime.

Interaction Characteristics

- All hardware involved communicates via standard **HTTP/HTTPS protocols**.
- No proprietary or special-purpose hardware is required.

4.2.5 Third-Party Hardware Assumptions

The system assumes:

- Users' hardware supports modern browsers.
- Administrator devices include a keyboard and mouse (for dashboard operations).

Hosting hardware remains available and functional to prevent downtime.

4.3 Software Interfaces

This section describes all software components that interact with the Digital Cake Ordering and Management System, including databases, operating systems, APIs, frameworks, libraries, and third-party services. It also identifies the types of data exchanged across these interfaces and the communication mechanisms involved.

4.3.1 Backend Software Interface – Spring Boot

Technology

- **Spring Boot Version:** 3.x
- **Java Version:** 17+
- **Build Tool:** Maven or Gradle

Purpose

- Handles business logic
- Processes REST API requests
- Manages authentication and authorization
- Communicates with PostgreSQL database through JPA/Hibernate

Data Exchanged

- **Incoming:**
  - JSON inputs from React (order details, customization data, admin actions)
- **Outgoing:**
  - JSON responses (catalogue data, order confirmation, order status, error messages)

Communication Method

- RESTful API over **HTTP/HTTPS**
- Controller → Service → Repository architecture

4.3.2 Frontend Software Interface – React.js

Technology

- **React Version:** 18+
- **Node.js Version:** 16+
- **Package Manager:** npm or yarn

Purpose

- Provides the UI for customers and administrators
- Sends API requests to the Spring Boot backend
- Renders dynamic content (catalogue, order tracking, admin dashboard)

Data Exchanged

- Sends:
  - Order details
  - Customization inputs

    - Admin updates (inventory, products, status changes)
- Receives:
    - Cake catalogue
    - Product images
    - Order status updates
    - Authentication tokens

Communication Method

- Axios/Fetch API (JSON-based REST communication)

4.3.3 Database Software Interface – PostgreSQL

Technology

- **PostgreSQL Version:** 13+
- **JDBC Driver:** PostgreSQL JDBC 42.x
- **ORM:** Hibernate/JPA

Purpose

- Stores structured relational data for:
    - Customers
    - Orders
    - Products
    - Inventory
    - Reports

Data Exchanged

- System stores and retrieves data via JPA repositories.
- Communication occurs through:
    - SQL queries generated by ORM
    - JDBC database connections

Communication Method

- TCP/IP over **Port 5432**
- Authentication required (username/password)

4.3.4 Node.js Interface (Build Tools & Package Management)

Technology

- **Node.js Version:** 16+
- **npm Version:** 8+

Purpose

- Builds and bundles frontend React application
- Manages third-party JavaScript dependencies
- Runs development server for UI testing

Data Exchanged

- No runtime data exchange with the backend
- Used only during development or frontend deployment build process

4.3.5 Authentication Interface – JWT (JSON Web Token)

Technology

- Spring Security
- JWT token library

Purpose

- Provides secure authentication for admin users
- Ensures only authorized users access admin panels

Data Exchanged

- **Incoming:** Login credentials
- **Outgoing:** JWT token

Communication Method

- Token added to API request headers:
  `Authorization: Bearer <token>`

4.3.6 Email Interface (Optional)

Technology

- SMTP service (Gmail SMTP, SendGrid, etc.)

Purpose

- Sends order confirmations or notifications

Data Exchanged

- Outgoing:
    - Customer email
    - Order ID
    - Order summary

4.3.7 File Storage Interface (Optional)

Technology

- Local file server OR cloud storage (e.g., AWS S3)

Purpose

- Stores cake images or downloadable reports

Communication Method

- API-based upload/download for cloud storage
- Direct file path access for local storage

4.3.8 External API Interfaces (Optional)

If implemented, the system may integrate with:

Payment Gateway APIs

- Accepts - Payment request JSON
- Returns - Payment confirmation/failure

Map/Location APIs

- Used for delivery cost or route calculation

These APIs require:

- API keys
- HTTPS communication
- Defined request/response JSON structures

4.3.9 Shared Data Between Components

The following data entities are shared across system components:

Data Entity	Shared Between	Purpose
Customer Data	React ↔ Spring Boot ↔ PostgreSQL	Order placement, tracking
Order Data	React ↔ Spring Boot ↔ PostgreSQL	Order management
Product & Cake Data	Spring Boot ↔ PostgreSQL ↔ React	Catalogue display
Inventory Data	Spring Boot ↔ PostgreSQL	Admin inventory updates
Authentication Tokens	React ↔ Spring Boot	Secure login

Data sharing follows the constraints of:

- JSON format (frontend ↔ backend)
- JPA/Hibernate entity mapping (backend ↔ database)

4.4 Communications Interfaces

This section describes all communication mechanisms and protocols used by the Digital Cake Ordering and Management System. It includes browser communication, backend-server interactions, database connections, email transmission (optional), and any security or synchronization requirements.

4.4.1 Web Communication Interface (Frontend ↔ Backend)

Protocol

- All communication between the React frontend and Spring Boot backend shall occur over: **HTTPS using HTTP/1.1 or HTTP/2**

Message Format

- All requests and responses must use **JSON** format.
- API endpoints must return appropriate HTTP status codes:
  - **200** – Success
  - **201** – Created
  - **400** – Bad Request
  - **401** – Unauthorized
  - **404** – Not Found
  - **500** – Server Error

Communication Method

- Frontend sends HTTP requests using:
  - Axios
  - Fetch API

Data Transfer Requirements

- Communication must use encrypted packets (TLS 1.2+).
- Typical payload size ranges from 1 KB to 200 KB.

4.4.2 Backend ↔ Database Communication (Spring Boot ↔ PostgreSQL)

Protocol

- Communication occurs using:
  - **TCP/IP**
  - Default PostgreSQL port **5432**

Message Format

- SQL queries (generated by Hibernate/JPA)
- Binary or text-based PostgreSQL protocol messages

Requirements

- Database credentials (username/password) are required.
- Data encryption at rest is recommended but optional for academic deployment.

4.4.3 Admin Authentication Interface

Protocol

- Backend uses **JWT (JSON Web Token)** for secure authentication.

Message Format

- Tokens are transmitted via HTTP headers:
  `Authorization: Bearer <token>`

Security Requirements

- Tokens must be:

  - Signed with a secure secret key
  - Transmitted only over HTTPS

4.4.4 Email Communication Interface (Optional)

If email notifications (order confirmation, password reset) are implemented:

Protocol

- SMTP protocol through a mail server such as:
  - Gmail SMTP
  - SendGrid
  - Mailgun

Message Format

- Plain-text or HTML-formatted emails
- JSON/Payload requests if using an email API (e.g., SendGrid API)

Security

- SMTP over TLS (Port 587)
- Passwords/API keys must remain encrypted in server environment variables

4.4.5 File Upload/Download Communication (Optional)

If storing cake images or downloadable reports:

Protocol

- HTTP/HTTPS
- Cloud storage APIs (AWS S3, Firebase Storage)

Message Format

- Multipart/form-data for image upload
- Binary data for file download

4.4.6 Browser Requirements (Client-Side Communication)

Communication with the user's browser must support:

- HTML5
- CSS3
- JavaScript ES6

- AJAX/Fetch requests
- Cookies (for basic session tracking if required)

Supported Browsers:

- Chrome
- Firefox
- Safari
- Edge

All must support TLS/SSL for secure communication.

4.4.7 Performance & Synchronization Requirements

Performance

- Average API response time must be under **2 seconds** (depending on hosting).
- High-resolution images should use optimized compression to reduce network load.

Synchronization

- Admin updates to product catalogue or inventory must reflect on the customer UI within:
  - **0–30 seconds**, depending on API response and caching.

Data Refresh

- Frontend may use:
  - On-demand fetch (user actions)
  - Periodic polling (optional)
  - WebSocket (optional future enhancement)

4.4.8 Security Requirements

Encryption

- All external communication must use **HTTPS/TLS 1.2+**.
- No plain HTTP connections allowed in production.

Data Integrity

- JWT tokens must include expiration and signature validation.
- Sensitive data (passwords, tokens) must never be transmitted or logged in plain text.

Firewall/Network

- The database port (5432) must only be accessible to the backend server.
- Admin dashboard must be accessible only after authentication.

5. Other Non-functional Requirements

5.1 Performance Requirements

This section defines the expected performance of the Digital Cake Ordering and Management System under normal and peak usage conditions. All performance requirements are measurable, verifiable, and aligned with the system's functional goals.

1. **System Response Time**
   The system should load all major pages (home, cake catalog, order form, admin dashboard) within **2–3 seconds** under normal internet conditions.
2. **Concurrent User Support**
   The system must support at least **20 simultaneous users** (customers + admin) without significant delays or performance degradation, suitable for a small/medium bakery.
3. **Order Processing Speed**
   Submitting an order must take **less than 2 seconds**, including validation, saving to the database, and generating an order ID.
4. **Database Query Performance**
   Common queries (fetching cakes, retrieving orders, loading inventory) must execute within **1 second** under normal load.
5. **System Availability**
   The application must maintain **99% uptime** during bakery operating hours (9 AM – 10 PM), ensuring uninterrupted customer ordering and admin operations.
6. **Scalability Provision**
   The system should allow performance improvements (e.g., caching, load balancing, optimizing queries) if user volume or order frequency increases in the future.
7. **Resource Usage Efficiency**
   The application must efficiently use server resources; CPU and memory consumption should remain stable during normal operations without causing system crashes or slowdowns.

5.2 Safety Requirements

Although the Digital Cake Ordering and Management System does not directly control physical machinery or life-critical functions, certain safety-related requirements must be addressed to prevent loss of data, financial loss, misuse, or harm resulting from incorrect system behavior.

The following safety requirements ensure the protection of user data, prevent accidental loss of information, and avoid situations that may disrupt bakery operations.

1. **Data Loss Prevention**
   The system must prevent accidental deletion of important records and enforce confirmation prompts before deleting orders, products, or inventory items.
2. **Safe Input Validation**
   All user inputs (order details, dates, quantities, custom messages) must be validated to prevent invalid or harmful data from entering the system.

3. **System Recovery Safety**
   In case of a system crash or power failure, all confirmed orders must remain in the database without duplication or corruption.
4. **Unauthorized Action Prevention**
   Only authorized users can access administrative functions such as product editing, inventory updates, and order status changes.
5. **Secure Error Handling**
   The system must not display technical system errors to users. Error messages should be safe, clear, and non-technical.

5.3 Security Requirements

1. **User Authentication & Role-Based Access Control (RBAC)**
   - The system shall require all administrators to log in using a secure authentication mechanism.
   - Only authorized users (admins) may access privileged sections such as product management, order status updates, and inventory controls.
   - Customers may access only their own order details, not administrative features.
2. **Secure Data Transmission (HTTPS/TLS)**
   - All communication between the client and server shall be encrypted using HTTPS/TLS protocols.
   - No sensitive information (passwords, user data, order details) may be transmitted in plain text.
3. **Password Protection & Storage Requirements**
   - User passwords shall be stored using salted and hashed encryption (e.g., bcrypt, Argon2).
   - Passwords must never be stored in plain text.
   - The system shall enforce minimum password strength guidelines (e.g., 8 characters, mixed types).
4. **Protection Against Common Web Attacks**
   - The system must defend against OWASP Top 10 vulnerabilities, including but not limited to:
     - SQL Injection
     - Cross-Site Scripting (XSS)
     - Cross-Site Request Forgery (CSRF)
     - Broken Authentication
     - Insecure Direct Object References (IDOR)
   - Server-side validation must be applied for all inputs.
5. **Data Privacy & Confidentiality Requirements**
   - User personal data (name, phone, address) must be accessible only to authorized system users.
   - The system shall comply with general privacy best practices, ensuring no sharing of user data with third parties.
   - Error logs must not contain sensitive user information.

5.4 Software Quality Attributes

1. **Usability**
   The system shall provide an intuitive and user-friendly interface that enables customers to browse and order cakes with minimal learning effort. Navigation must be simple, and all-important actions should be clearly labeled.
2. **Reliability**
   The system must maintain consistent performance and accuracy in order processing, ensuring that order data is not lost, duplicated, or corrupted. The system should achieve at least **99% uptime** during normal operating hours.
3. **Performance Efficiency**
   Pages should load within **2–3 seconds** under normal network conditions, and the system must support at least **20 simultaneous users** without degradation in performance (for academic scope).
4. **Maintainability**
   The system shall be structured using modular, well-documented code to allow developers to easily update features, fix bugs, and add new components. Clear separation between frontend, backend, and database layers must be maintained.
5. **Portability**
   The application shall run smoothly on common web browsers (Chrome, Firefox, Edge) and be accessible across multiple device types including PCs, laptops, and mobile devices without requiring any additional installation.
6. **Scalability**
   The system should be designed so that additional features (e.g., payment gateway, delivery tracking, multi-branch support) can be added in future releases without major architectural changes.
7. **Interoperability**
   The backend must be capable of integrating with third-party services in the future (e.g., SMS notifications, payment systems, email services) using standard RESTful APIs.
8. **Correctness**
   The system must process all orders, calculations, and inventory updates accurately, ensuring that no discrepancies occur between customer orders and admin records.
9. **Robustness**
   The system should handle invalid inputs, missing data, and system errors gracefully without crashing or exposing sensitive internal information.
10. **Testability**
    The system architecture must allow unit testing, integration testing, and user acceptance testing. All major modules should provide clear inputs and outputs for validation.

6. Other Requirements

1. **Database Requirements**
   - The system must use **PostgreSQL** as the primary database.
   - All tables must maintain referential integrity using foreign keys.
   - Data types must be selected to ensure accuracy (e.g., numeric for prices, text for descriptions, timestamp for dates).
   - Database must support daily backups and basic recovery in case of data loss.
2. **Internationalization Requirements (Optional)**
   - The system should support English as the primary language.
   - The design should allow easy addition of other languages in the future (e.g., Sinhala, Tamil) without major architectural changes.
3. **Legal & Compliance Requirements**
   - The system must comply with general data protection best practices (no storing sensitive data unnecessarily).
   - User personal information may only be used for order processing and must not be shared externally.
   - Images, icons, and UI assets should either be original, royalty-free, or properly licensed.
4. **Reusability Requirements**
   - Components such as the order module, product module, and inventory module must be designed in a modular format so they can be reused in potential future projects or system expansions.
   - API endpoints must follow RESTful standards to enable reuse in mobile applications later.
5. **Logging & Monitoring Requirements**
   - The system must record key system events such as login attempts, order submissions, inventory updates, and admin actions.
   - Error logs must be stored securely and must not expose sensitive data.
   - Logs should assist developers in debugging without revealing internal system structure to end-users.
6. **Backup & Recovery Requirements**
   - Automatic backups should occur daily (based on deployment environment).
   - The system must provide procedures for recovery in case of accidental deletion, hardware failures, or database corruption.
7. **Future Expansion Requirements**
   The system must be built with the ability to integrate:
   - Online payment gateways
   - Delivery tracking modules
   - Multiple bakery branches
   - Loyalty programs or discount systems

   These features are not part of the current version but should be easily extendable.
8. **Accessibility Requirements**
   - The user interface should be accessible on both desktop and mobile devices.
   - Text and buttons must be readable and clickable for users with basic device skills.

Appendix A: Glossary

Term / Acronym	Definition
SRS	Software Requirements Specification – A document that describes all functional and non-functional requirements of the system.
System	The Digital Cake Ordering and Management System being developed.
Administrator (Admin)	A privileged user responsible for managing products, orders, inventory, and system settings.
Customer	An end user who browses cakes, customizes orders, and submits online orders.
Order	A confirmed request submitted by a customer for one or more cakes. Includes customization details, dates, and contact information.
Customization	The additional options selected by customers (size, flavor, icing, message, decorations).
Inventory	The list of ingredients and materials required for cake production, tracked by the admin.
UI (User Interface)	The graphical layout through which customers and admins interact with the system.
Backend	The server-side logic built using Spring Boot (or Node.js if applicable), responsible for processing data, API operations, and database communication.
Frontend	The client-side interface built using React, responsible for displaying pages and collecting user inputs.
API	Application Programming Interface – The communication layer that allows the frontend and backend to exchange data.
HTTP / HTTPS	Communication protocols used for transferring data between the client and server. HTTPS ensures encrypted communication.
REST	Representational State Transfer – A standard style for designing web APIs.
Database	The PostgreSQL storage system used for managing persistent data such as orders, cakes, users, and inventory.
Entity	A real-world object or concept represented in the database (e.g., Cake, Order, Customer).
ERD	Entity-Relationship Diagram used to visually represent database entities and their relationships.
DFD	Data Flow Diagram showing how data moves through the system from input to output.
Class Diagram	A UML diagram showing system classes, attributes, and relationships for system design.
Authentication	Verifying the identity of a user (e.g., login process).
Authorization	Determining what actions a user is permitted to perform based on their role.
Session	A temporary connection between the user and the system during which user actions are tracked.
Validation	The process of checking that user input is correct and acceptable before saving data.
Deployment	The process of installing and running the system on a server or hosting environment.

Appendix B: Analysis Models

This appendix includes key analysis models used to understand, design, and visualize the overall behavior and structure of the Digital Cake Ordering and Management System. The following diagrams represent how data flows through the system, relationships between database entities, structural class relationships, and order status transitions.

B.1 Data Flow Diagram (Level 0 – Context Diagram)

Figure 3 Data Flow Diagram (Level 0 – Context Diagram)

B.2 Data Flow Diagram (Level 1 – High-Level Processes)

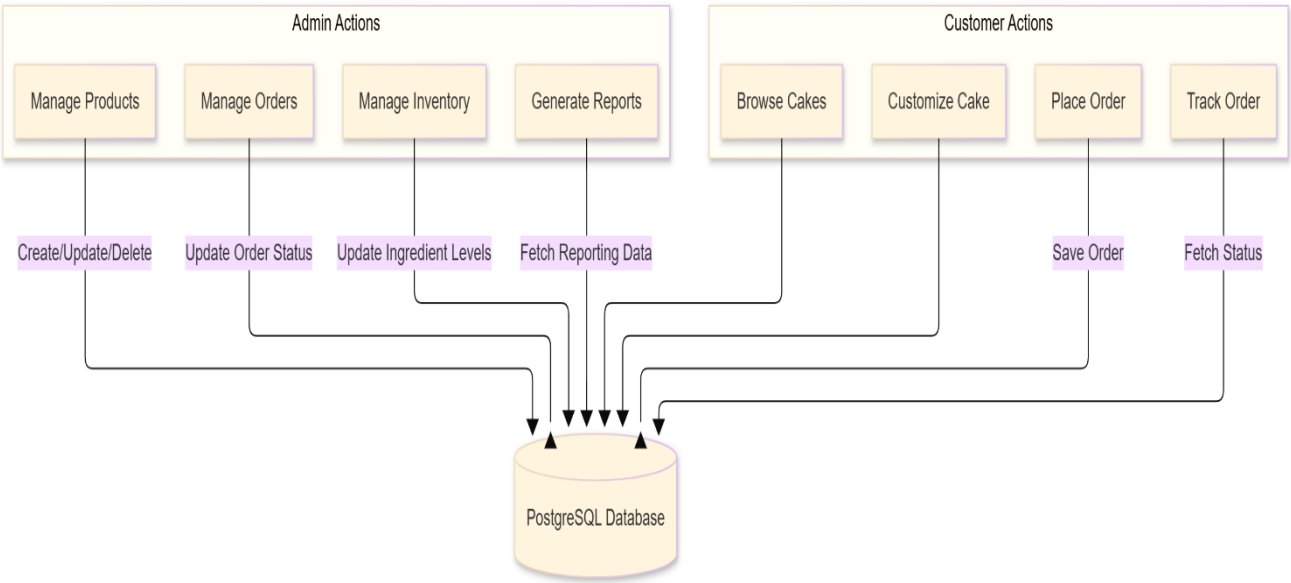

Figure 4 B.2 Data Flow Diagram (Level 1 – High-Level Processes)

B.3 Entity–Relationship Diagram (ERD)

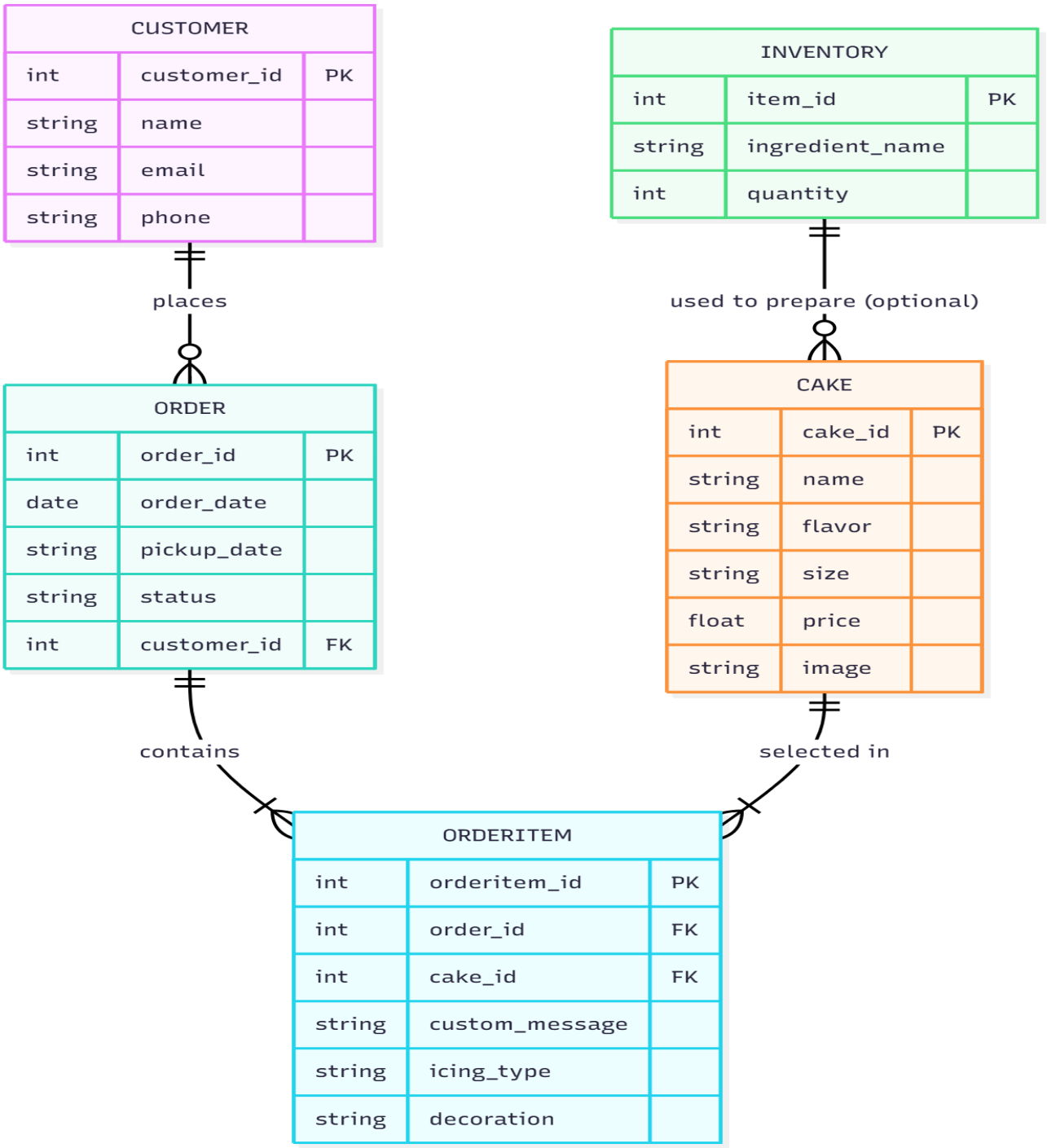

Figure 5 B.3 Entity–Relationship Diagram (ERD)

B.4 State Transition Diagram (Order Status Flow)

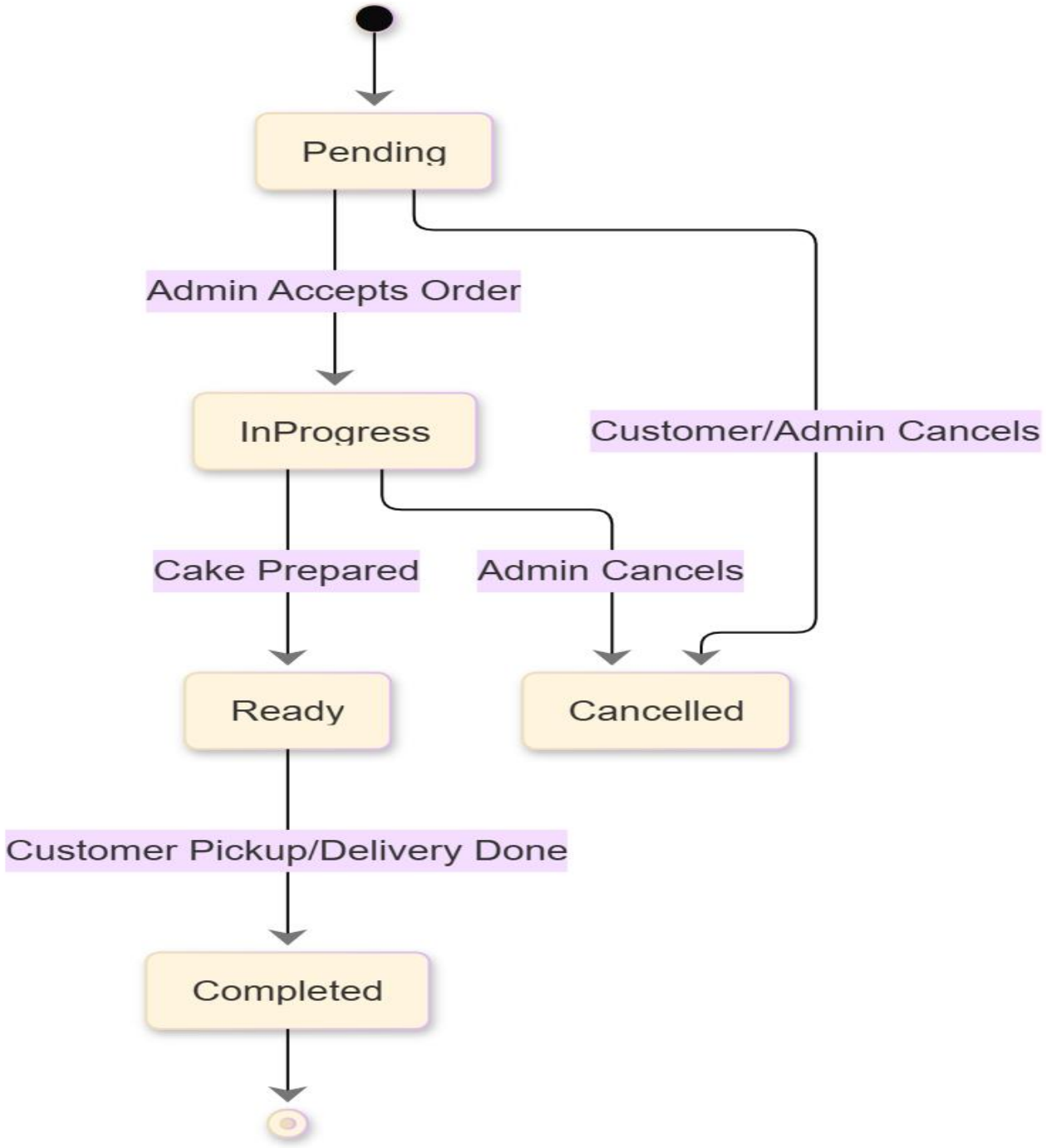

Figure 6 B.5 State Transition Diagram (Order Status Flow)

B.5 Sequence Diagram (Order Placement)

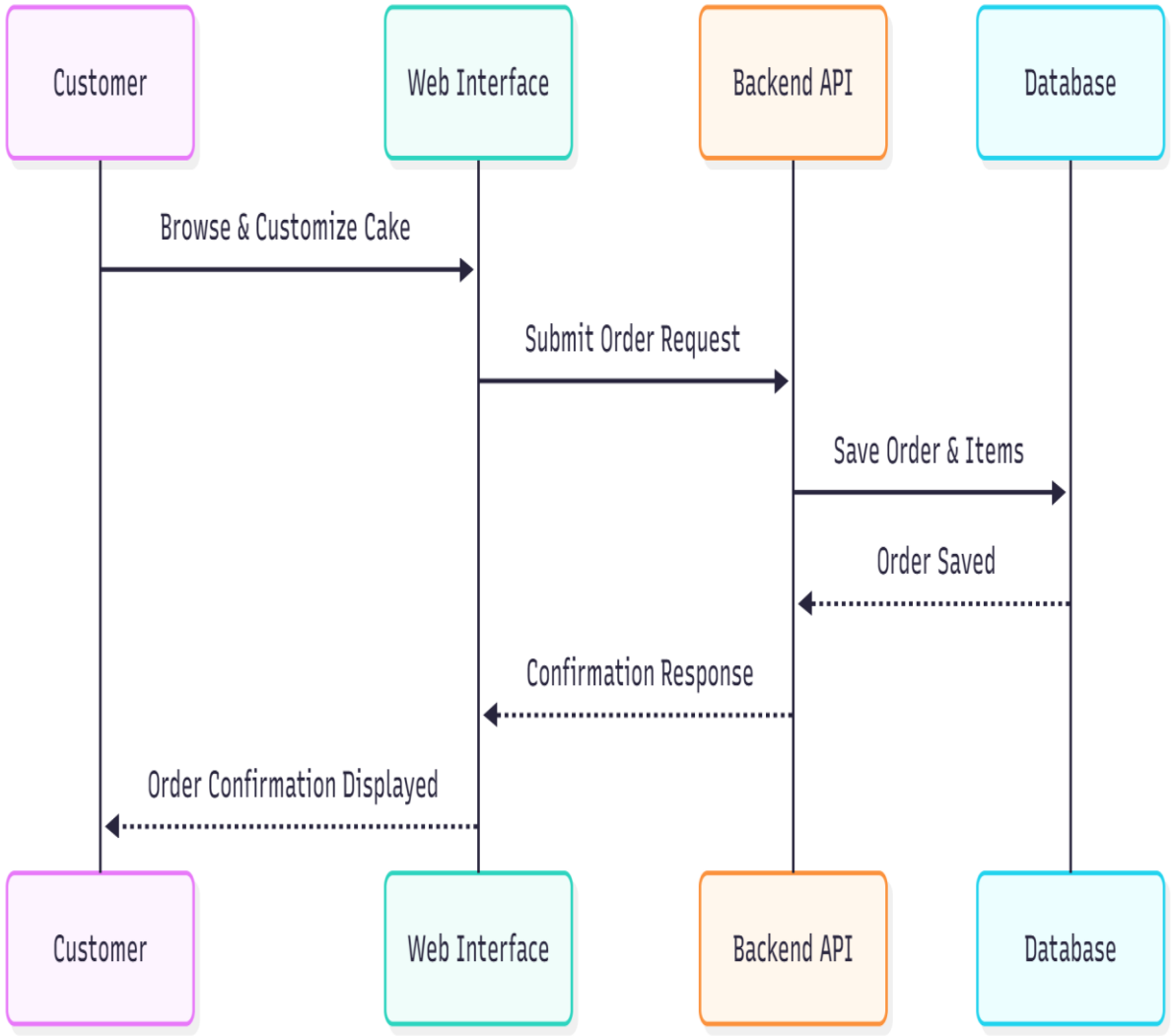

Figure 7 B.5 Sequence Diagram (Order Placement)

Appendix C: Issues List

This section lists all unresolved issues, pending decisions, and items that require clarification or further analysis. These issues may affect the final system design, functionality, or implementation.

Issue ID	Description of Issue	Impact on System	Status	Notes / Required Action
ISS-01	Decision on whether to include online payment integration (e.g., card payments, digital wallets).	Affects order workflow, security, backend integration, database structure.	Pending	TBD based on system scope or future enhancement plan.
ISS-02	Requirement for multi-language support (Sinhala/Tamil).	Affects UI design, content storage, React component structure.	Under Discussion	Need confirmation whether single-language version is sufficient.
ISS-03	Automatic inventory deduction during order placement.	Impacts inventory accuracy and cake production workflow.	Pending	Need to confirm if deduction should be manual or automatic.
ISS-04	Real-time order status notifications (SMS/Email).	Affects integration with external services and adds cost/complexity.	Not Decided	Decide if academic scope includes real-time notifications.
ISS-05	Admin role expansion (should multiple admin levels exist?).	Impacts authentication model, RBAC rules, backend logic.	Pending	Need confirmation whether to add Manager / Staff roles.
ISS-06	Reporting Module scope (basic or detailed analytics).	Changes backend queries, database indexing, UI requirements.	Under Review	Must choose between minimal reports or full analytics.
ISS-07	Handling custom cake designs with image uploads.	Affects storage requirements, database fields, file upload security.	TBD	Decide if customers can upload design reference images.
ISS-08	Deployment environment (local, cloud hosting, or VPS).	Determines performance, security, and scalability.	Not Finalized	Need to confirm final hosting setup.
ISS-09	Mobile responsiveness vs. dedicated mobile app.	Affects UI frameworks and design time.	Pending	Confirm if responsive design alone is sufficient.
ISS-10	Data retention duration for past orders and logs.	Impacts database storage, privacy compliance, backup strategy.	TBD	Requires decision from the project owner (you).